

**SYLLABUS :-**

Properties of scalars, vectors and tensors; Elements of vector analysis; Gradient, Divergence and Curl, Gauss's Divergence Theorem, Stokes Theorem, Definition of field, Classification of Fields, Gravitational field, Newton's Law of Gravitation, Gravitation potential and fields due to bodies of different geometric shapes, Electrostatic fields, Coulomb's law, Electrical permittivity and dielectric constant; Basic guiding equations; Magnetostatic Field, Origin of the Magnetic Field, Ampere's Law, Biot and Savart's Law, Geomagnetic Fields, Magnetic Fields due to different type of structures, Solution of Laplace equation in Cartesian Coordinate, Cylindrical Polar Coordinate and Spherical Polar Coordinate, Complex Variables in Potential Theory, Green's Theorem in Potential Theory. Concept of image in Potential Theory, Analytical Continuation of potential Fields, Numerical Methods in Potential Theory Electrical fields in geophysics, point source, continuous distribution and double layers, equipotential and line of force. Current and potential in the earth, basic concept and equations of electromagnetics, Maxwell's equations, boundary conditions, elliptic polarization, electromagnetic potential and waves, radiation from dipoles, retarded potential, near and far fields, radiation resistance, EM field of a loops of wire on half-space, multi-layered media, impedance and its application